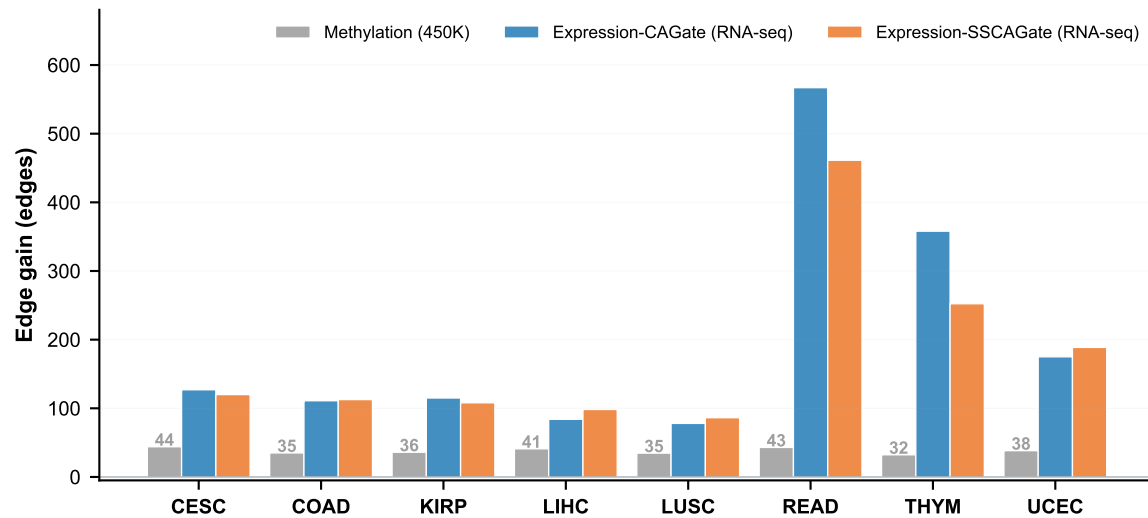
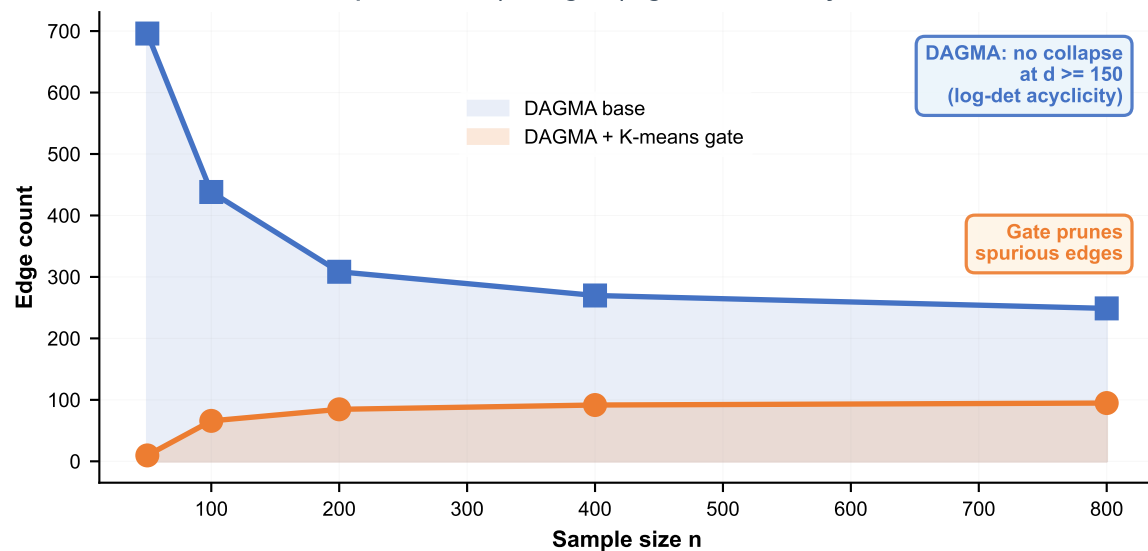


a | Modality-robust phase transition - methylation attenuated but consistent (lower CpG SNR)



b | DAGMA boundary condition: gate targets matrix-exponential weakness

Mechanism: trace-exp bottleneck (not log-det) - gate unnecessary for DAGMA - evidence FOR specificity, not failure



c | Computational cost - gate overhead negligible

